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About

This is a *sample* book written in **Markdown**. You can use anything that Pandoc's Markdown supports; for example, a math equation $a^2 + b^2 = c^2$.

0.1 Usage

Each **bookdown** chapter is an .Rmd file, and each .Rmd file can contain one (and only one) chapter. A chapter *must* start with a first-level heading: `# A good chapter`, and can contain one (and only one) first-level heading.

Use second-level and higher headings within chapters like: `## A short section` or `### An even shorter section`.

The `index.Rmd` file is required, and is also your first book chapter. It will be the homepage when you render the book.

0.2 Render book

You can render the HTML version of this example book without changing anything:

1. Find the **Build** pane in the RStudio IDE, and

2. Click on **Build Book**, then select your output format, or select "All formats" if you'd like to use multiple formats from the same book source files.

Or build the book from the R console:

```
bookdown::render_book()
```

To render this example to PDF as a `bookdown::pdf_book`, you'll need to install XeLaTeX. You are recommended to install TinyTeX (which includes XeLaTeX): https://yihui.org/tinytex/.

0.3 Preview book

As you work, you may start a local server to live preview this HTML book. This preview will update as you edit the book when you save individual .Rmd files. You can start the server in a work session by using the RStudio add-in "Preview book", or from the R console:

```
bookdown::serve_book()
```

1
Mathematical modelling

Maths is used to make sense of the world around us, and help us to make decisions, in our personal lives and in the worlds of business, politics, science, and so on. This could involve, for example:

- Creating and solving equations.
- Plotting graphs.
- Making calculations.
- Using computer software.

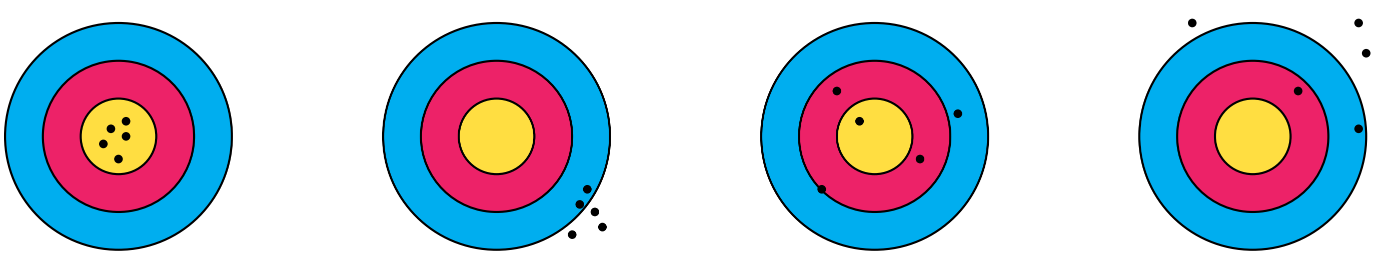
Each of these things could be thought of as creating a **mathematical model**. This chapter will introduce and explore a range of different kinds of model that may be useful, and when they may be encountered in the real world.

> Quote
>
> *"All models are wrong, but some are useful."*
> George Box, Statistician

The above quote highlights that mathematical models don't always need to be able to produce 'the *right* answer', and in many situations such a thing is not even possible. Instead, the goal is often a simply a *useful* answer. Since we are entering a world of often *not knowing the true answer*, it is helpful to be learn some related vocabulary:

- **Accuracy** relates to how much a measurement or process leads towards the *true value*.
- **Precision** relates to how close repeated measurements or processes are to *each other*.

The results of four different archers aiming for the centre of a target should help illustrate the difference:



Accurate ✓	Accurate ✗	Accurate ✓	Accurate ✗
Precise ✓	Precise ✓	Precise ✗	Precise ✗

1.1 Systematic Estimation

The 20th century physicist Enrico Fermi was known for performing 'back of an envelope' calculations which often gave surprisingly reliable and *accurate* results. Calculations of this type are therefore sometimes called *'Fermi estimates'*. The goal of these *systematic estimates* is to quickly get a sense of the *scale* of numerical answers to complex or even impossible questions such as:

Questions

- *How many times might a teacher take a class register in their career?*
- *How much on average does a large supermarket spend on staff wages each week?*
- *How many miles are driven by cars on Scottish roads in a month?*

These calculations might involve using a mixture of *approximated*, *estimated* and *exact* numbers. For example:

- There are **exactly** 60 seconds in one minute.
- There are **approximately** 30 days on average in a month.
- We might **estimate** the mass of a typical family car to be 1000 kilograms.

It is important to communicate clearly any **assumptions** made in reaching your answer, such as the *estimation* above.

Example 1.1A

Problem:
A secondary school in a city centre in Scotland has approximately 800 pupils.
Estimate the total amount spent on lunches in the school's dining hall per month.

Solution:
Assume that:

- Half of the pupils in the school buy their lunch from the dining hall.
- On average £4 is spent on lunch each day by those pupils.

$$400 \times 4 \times 20 = 32000 \approx \text{£}30{,}000$$

You may have may have estimated a different proportion of pupils that use the dining hall, and a different average spend. It is expected your final answer will vary based on your estimates of the different *variables*.

Example 1.1B

Problem:
Estimate the total volume of water that a typical person drinks during their childhood.

Solution:
Assume that:

- 'Childhood' lasts 18 years (from 0 to 18).
- On average a child might drink 1 litre of water a day.

$$1 \times 365 \times 18 = 6570 \approx 6600 \text{ litres}$$

1.2 Modelling Relationships

We may wish to explore the *relationships* between two or more variables, such as when considering:

Questions

- How will the **money** in my savings account grow over time if I make regular payments?
- What might a car's **braking distance** by when travelling at different speeds?
- How might the distance travelled in a taxi be related to the **fare**?

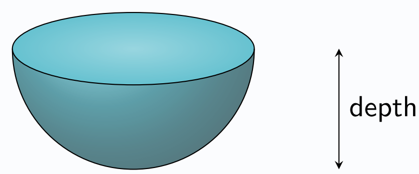
We should aim to identify where possible for any relationship:

- The independent variable - commonly either *time* or one which we control.
- The **depdendent** variable - which changes in **response** to the other.

Initially a consideration of a possible relationship should be *descriptive*, thinking about when the dependent variable might be increasing slowly, increasing more quickly, decreasing, at zero or other specific values. *Graphs* can help:

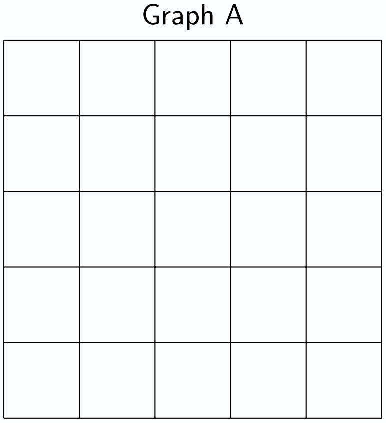
Example 1.2

Problem:
Bob pours coffee into an empty 200ml capacity coffee cup at a constant rate.
The interior of the coffee cup is in the shape of a hemisphere, as shown below.

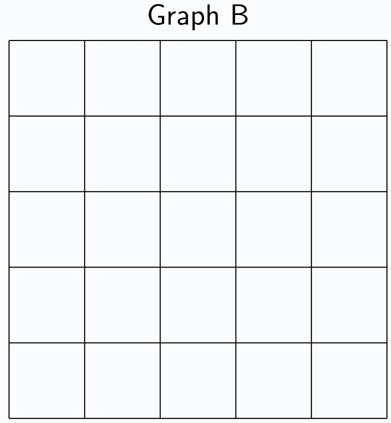


State which of graphs A, B or C below might be used to model the relationship between the time since Bob began pouring the coffee and the depth of coffee in the cup, explaining your answer.

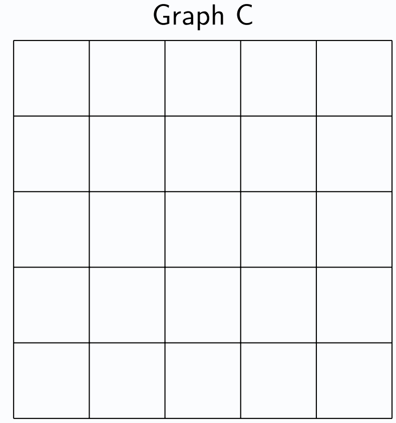
Graph A



Graph B



Graph C



Solution:
Graph ? since *blah blah blah*...

1.3 Linear Relationships